

ME231 Statics Notes

1 Lecture — 24/09/2026

Curriculum Position

- Week-1 — Course intro, syllabus, Newton's laws, vectors.

Key Content

1. Mechanics
2. Basic Quantities
3. Idealizations and an intro into rigid body mechanics
4. Newton's Laws of Motion
5. Newton's Law of Gravitational Attraction
6. SI Units
7. Unit Prefixes and Rules for Unit Prefixes
8. Dimensional Homogeneity
9. Sig-Figs and Rounding
10. General Procedure for Analyzing Systems

Instructor Emphasis

- Use 3 Sig-Figs for calculations.
- Take $g = 9.81 \text{ m/s}^2$ for calculations.
- He cares a lot about unit prefix rules.
- It is recommended that calculations be done with the units alongside, so that the resulting unit gets verified.
- Retain a greater number of digits than the problem data for calculations.
- For solving problems with different units, converting to base units first is recommended.

Announcements

1. Email appointment before office visit.
2. Lecture notes are provided and said to be adequate.
3. Nonattendance allowed up to 4 days.
4. AA is (100-90) by default, Grading curve is possible based on performance.
5. Worked out problems will be provided on the course Teams group.
6. 1 Midterm (35%), 1 Final (45%), Pop-Quizzes (5% each) Announced a week before.
7. Textbook: Hibbeler, R.C. Engineering Mechanics
8. Next week will consist of 3 sessions.

Open Questions

Independent Follow-up

- Minimum: Have a light read.
- Exploration: Watch some stuff on statics and FEA

2 Lecture — 01/10/26

Curriculum Position

- Week-2 — Forces, Vectors, and Vector Operations

Key Content

1. Vector Addition of Forces
2. Cartesian Vector Notation and the Decomposition of Vectors
3. Right Handed Coordinate Systems and 3D Vector Decomposition
4. Unit Vectors and the Direction of a Cartesian Vector
5. Concurrent Force Systems and x, y, z Coordinates
6. Position Vectors and Force Vectors Directed Along a Line
7. Dot Products, Cartesian Vector Formulation

Instructor Emphasis

- Working with units while solving problems is highly encouraged for unit consistency and orders of magnitude assurance.
- The instructor consistently preferred the laws of sines, cosines, and parallelograms over vector decomposition when it came to simpler problems.
- Vector decomposition is advised for the addition of more than 2 vectors.
- Studying example 2.17 is recommended.

Announcements

Open Questions

Independent Follow-up

- Minimum: Work on the vector addition examples provided in the lecture notes.
- Exploration: Watch some stuff on statics and FEA.

3 Lecture — 08/10/26

Curriculum Position

- Week-3 — Free Body Diagrams and Moments

Key Content

1. Free Body Diagrams
2. Springs
3. Cables and Pulleys
4. The Procedure of Drawing FBD's
5. Coplanar Systems and Simple Force Equilibria
6. The Procedure for Analysis
7. Three Dimensional Force Systems
8. Moment of a Force — Scalar Formation
9. Cross Products, Laws of Operations
10. Principle of Transmissibility
11. Varignon's Theorem

Instructor Emphasis

- Follow the lecture notes closely for FBD drawing conventions.
- Use over-arrows to specify vector operations.
- Provide angle related answers in degrees only.
- Use cross products and vector equations in 3D problems for simplicity.

Announcements

1. Content scheduled for the next week started this week.

Open Questions

Independent Follow-up

- Review the lecture notes, go over relevant problems.